₹14,500 crore boost for energy storage projects

NANDINI KESHARI
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The Union Cabinet on Wednesday approved a ₹14,500-crore Viability Gap Funding (VGF) scheme to support the development of 92 gigawatt-hours (Gwh) of grid-scale energy storage capacity over FY27-FY31, as the government seeks to strengthen the power grid and facilitate greater integration of renewable energy.

The scheme will support new capacity equivalent to 170 per cent of the country's existing grid-scale energy storage capacity and help meet its estimated requirement of 411 Gwh by 2031-32, the Cabinet said.

Of the 92 Gwh to be developed, 60 Gwh will come from hydro pumped storage plants (PSPs), 30 Gwh from battery energy storage systems (BESS) and the remaining 2 Gwh from alternative energy storage technologies. The projects are expected to attract an estimated investment of ₹ 97,600 crore. The VGF support is intended to improve their financial viability and accelerate the deployment of large-scale energy storage infrastructure.

The scheme is aimed at strengthening India's power sector infrastructure, enabling large-scale integration of renewable energy and ensuring a reliable, affordable and round-the-clock power supply. It is also expected to give an impetus to domestic manufacturing of battery energy storage systems, reduce import dependence and enhance the country's long-term energy security, the Cabinet said.

Energy storage capacity will be allocated through a tariff-based competitive bidding process. Power distribution utilities will be eligible for VGF support for BESS and PSP projects, while projects based on alternative storage technologies will be implemented by public and private sector entities. The scheme also seeks to strengthen India's domestic energy storage manufacturing ecosystem through demand-side financial support and reduce dependence on imported components.

To support this objective, BESS projects will be subject to a minimum Domestic Content Requirement of 40 per cent.

The additional storage capacity is expected to facilitate greater utilisation of renewable energy by addressing fluctuations in generation. This would reduce dependence on conventional power generation to meet round-the-clock electricity demand, the Cabinet said.